

$$\textcircled{T_4} \quad] f \in R[a, b] \Rightarrow$$

$$\left| \int_a^b f dx \right| \leq \left| \int_a^b |f| dx \right|$$

$$\Delta \quad |f| \in R[a, b] \quad (\uparrow)$$

$$\Rightarrow \left| \sum_{i=1}^n f(\xi_i) \Delta x_i \right| \leq \left| \sum_{i=1}^n |f(\xi_i)| \Delta x_i \right|$$

$$\lim_{\lambda \rightarrow 0} \Rightarrow \lim_{\lambda \rightarrow 0} (\dots) \leq \lim_{\lambda \rightarrow 0} (\dots)$$

$$\left| \int_a^b f dx \right| \leq \left| \int_a^b |f| dx \right| \quad \Delta$$

$\textcircled{T_5}$ (о среднем)

$$\textcircled{1} \quad] f, g \in R[a, b]$$

$$\textcircled{2} \quad g - \text{знаковая } m[a, b]$$

$$] m = \inf_{[a, b]} f(x)$$

$$M = \sup_{[a, b]} f(x)$$

$\Rightarrow$

$$\exists \mu \in [m, M] :$$

$$\int_a^b f \cdot g dx = \mu \int_a^b g dx \quad (*)$$

$$\text{even } f \in C^0[a, b] \Rightarrow$$

$$\exists \xi \in [a, b] :$$

$$\int_a^b f \cdot g dx = f(\xi) \int_a^b g dx$$

$$\Delta \textcircled{\text{I}}] \quad g(x) > 0, \quad x \in [a, b] \Rightarrow$$

$$m \leq f(x) \leq M$$

$$m g(x) \leq f(x) g(x) \leq M g(x)$$

$$m \int_a^b g(x) dx \leq \int_a^b f(x) g(x) dx \leq M \int_a^b g(x) dx$$

$$\textcircled{1}] \int_a^b g(x) dx = 0 \Rightarrow \int_a^b f g dx = 0 \Rightarrow \textcircled{*} \text{ before}$$

$$\textcircled{2}] \int_a^b g(x) dx \neq 0 \Rightarrow \int_a^b g(x) dx > 0$$

$\Rightarrow$

$$m \leq \frac{\int_a^b f \cdot g dx}{\int_a^b g dx} \leq M \Rightarrow \textcircled{*} \text{ before}$$

$$\textcircled{\text{II}}] f \in C^0[a, b] \Rightarrow \text{no } \textcircled{\text{I}} \text{ b-K } \forall \mu \in [m, M]$$

$$\exists \xi \in [a, b]: f(\xi) = \mu$$

$$\Rightarrow \int_a^b f \cdot g dx = f(\xi) \int_a^b g dx \quad \Delta$$

$$\text{Zun } \int_a^b f(x) dx = \mu(b-a) \quad \boxed{\int_a^b g(x) = 1}$$

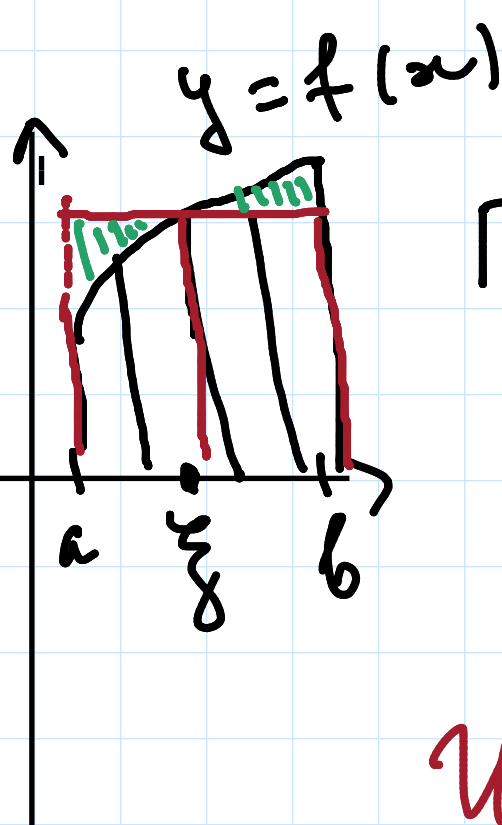
$$m < \mu < M$$

$$2) \int_a^b f(x) dx = f(\xi)(b-a)$$

$\Downarrow$

$$f(\xi) = \frac{1}{b-a} \int_a^b f(x) dx$$

cp. zu $f(x)$ in $[a, b]$



$$\Pi_{\text{тр}} = \int_a^b f \, dx$$

Интерпол с переменными
вершинами пределов.

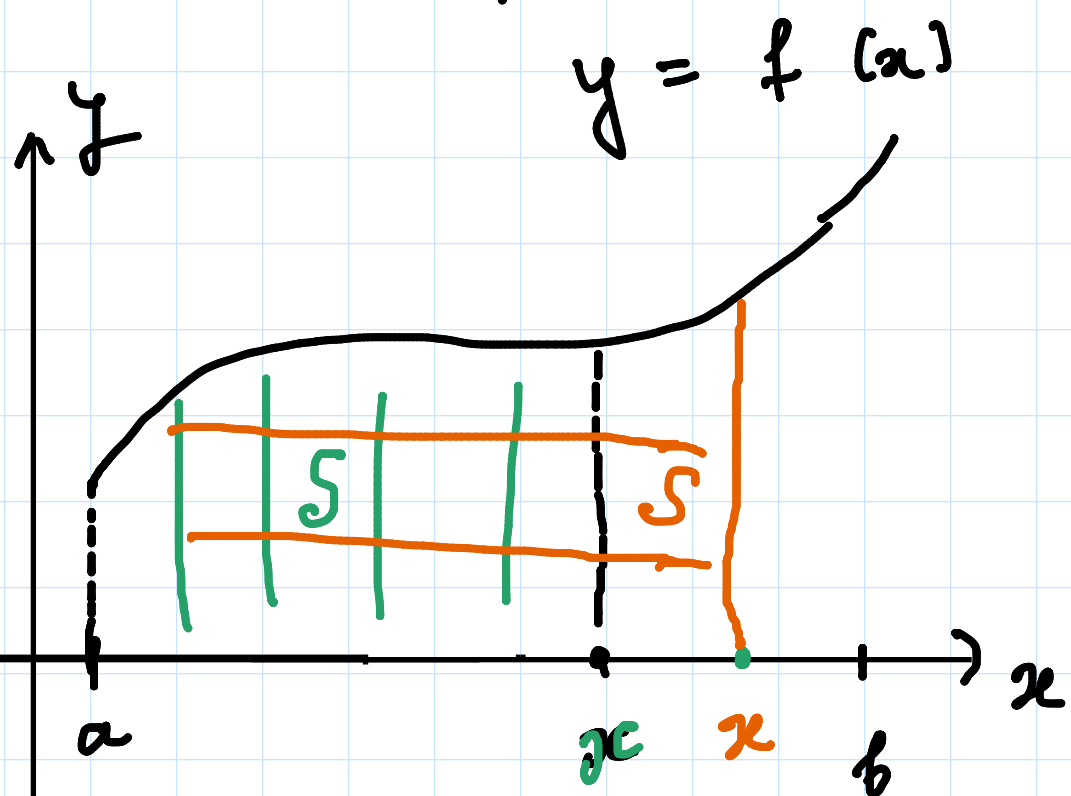
Опр $f \in R[a, b]$, $x \in [a, b]$

Функция

$$\Phi(x) = \int_a^x f(t) \, dt$$

наз. интерпол

с перем. в. пределом



$$\text{Зам } \Phi(x) = \int_a^x f(t) \, dt$$

$$\textcircled{T} \quad \Phi(x) \in C^0[a, b]$$

$$\Delta \quad] x_0 \in [a, b], \quad x_0 + \Delta x \in [a, b]$$

$$f \in R[a, b] \Rightarrow f - \text{ог на } [a, b]$$

$$\Rightarrow \exists C > 0 : |f(x)| \leq C, \quad x \in [a, b]$$

$$\circ \quad |\Delta \Phi| = |\Phi(x_0 + \Delta x) - \Phi(x_0)| =$$

$$= \left| \int_a^{x_0 + \Delta x} f(t) \, dt - \int_a^{x_0} f(t) \, dt \right| =$$

$$= \left| \int_{x_0}^{x_0 + \Delta x} f(t) \, dt \right| \leq \int_{x_0}^{x_0 + \Delta x} |f(t)| \, dt \leq C \Delta x \quad \textcircled{\leq}$$

$$\textcircled{\leq} C \left| \int_{x_0}^{x_0 + \Delta x} dt \right| = C |\Delta x|$$

$$\Rightarrow \lim_{\Delta x \rightarrow 0} |\Delta \Phi| \stackrel{\text{def}}{=} 0 \Rightarrow \Phi(x) \in C^0(x_0).$$

$$\text{т.к. } x_0 - \forall \text{ } \eta \in [a, b] \Rightarrow \Phi(x) \in C^0[a, b] \quad \Delta$$

$\textcircled{\text{T}}$ Φ — функция в точке непрерывна f

т.е. : если $f \in C^0(x_0)$, $x_0 \in [a, b] \Rightarrow$

$$\Phi(x) \in D(x) : \Phi'(x_0) = f(x_0)$$

$$\Delta \quad] f(x) \in C^0(x_0), [x_0 + \Delta x \in [a, b]$$

$$\Phi - \pi : \lim_{\Delta x \rightarrow 0} \frac{\Delta \Phi}{\Delta x} = f(x_0) \quad ?$$

$$\textcircled{?} \quad \left| \frac{\Phi(x_0 + \Delta x) - \Phi(x_0)}{\Delta x} - f(x_0) \right| =$$

$$= \left| \frac{\int_{x_0}^{x_0 + \Delta x} f(t) dt - \int_{x_0}^{x_0} f(t) dt}{\Delta x} - f(x_0) \right| =$$

$$= \left| \frac{\int_{x_0}^{x_0 + \Delta x} f(t) dt - f(x_0) \Delta x}{\Delta x} \right| = \left| \frac{\int_{x_0}^{x_0 + \Delta x} [f(t) - f(x_0)] dt}{\Delta x} \right| \leq$$

$$\leq \frac{\int_{x_0}^{x_0 + \Delta x} |f(t) - f(x_0)| dt}{|\Delta x|} \quad \textcircled{\leq}$$

$$] f \in C^0(x_0) \stackrel{\text{def}}{\Rightarrow} \forall \varepsilon > 0 \exists \delta(\varepsilon) > 0 :$$

$$\forall t \in [a, b] : |t - x_0| < \delta \Rightarrow |f(t) - f(x_0)| < \varepsilon$$

$$] |\Delta x| < \delta \quad \textcircled{=}$$

$$\textcircled{<} \quad \varepsilon \frac{\left| \int_{x_0}^{x_0 + \Delta x} dt \right|}{|\Delta x|} = \varepsilon \frac{|\Delta x|}{|\Delta x|} = \varepsilon$$

$$\Rightarrow \lim_{\Delta x \rightarrow 0} \frac{\Delta \Phi}{\Delta x} = f(x_0) \quad \Delta$$

Следствие $\forall f(x) \in C^0[a, b]$

$$\exists \text{ непрерывная } F(x) = \int_a^x f(t) dt + C = \Phi(x) + C$$

$$\begin{aligned} \Delta \exists f(x) \in C^0[a, b] &\Rightarrow \exists \Phi(x) = \int_a^x f(t) dt : \\ \exists \Phi'(x) &= f(x) \Rightarrow \Phi(x) - \text{непрерывная } f \text{ на } [a, b] \\ &\Rightarrow F(x) = \Phi(x) + C \quad \Delta \end{aligned}$$